Configuration Manual

MSc Research Project
Masters in Artificial Intelligence

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MSc Project Submission Sheet

School of Computing

Student Name:	Priyanka Bellamkonda
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Configuration Manual

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1 Development Environment Setup

All experiments and implementation were conducted using **Google Colab Pro+**, a cloud-based Jupyter environment with GPU acceleration. The environment supports direct use of Python 3.10 and comes pre-installed with most machine learning libraries.

To replicate this project locally, you can use:

- **Python version**: 3.10.12
- **Notebook Environment**: Jupyter Notebook (via Anaconda or VS Code)
- **GPU Acceleration** (optional for TabNet): NVIDIA Tesla T4 / Colab's GPU backend

Required packages were installed using pip commands. Use the following to install all core dependencies:

"pip install pandas numpy matplotlib seaborn scikit-learn xgboost pytorch-tabnet"

2 Dataset and Data Preparation

The dataset used was the **H&M Personalized Fashion Recommendation Dataset** publicly available on Kaggle.

Three files were used:

- transactions_train.csv – Transaction records
- articles.csv – Product metadata
- customers.csv – Customer profiles

All three files, must be placed in the same working directory and loaded using pandas.read_csv().

Preprocessing steps included:

- Merging datasets on article_id and customer_id
- Sampling 2 million purchase transactions (positive class)
- Generating 2 million synthetic non-purchase entries (negative class) with realistic transaction features
- Handling missing values using median/mode
- Encoding categorical variables using LabelEncoder
- Splitting data using stratified 80:20 train-test split

No additional cleaning scripts are required beyond what is present in the main notebook.

3 Model Training and Execution

Two models were implemented and trained:

a. XGBoost

- Model: XGBClassifier from the xgboost library
- Training: Performed in less than 60 seconds using CPU
- Outputs: Model file, confusion matrix, classification report, ROC curve

b. TabNet

- Model: TabNetClassifier from pytorch-tabnet
- Training: Conducted on GPU over 10 epochs
- Outputs: Attention masks, performance metrics, ROC curve

Training, evaluation, and visualizations are all done in the notebook. Models were saved using joblib and PyTorch checkpointing.

4 Deployed Demo URL

The trained model is served in Hugging Face Spaces in the form of a web application, at: https://huggingface.co/spaces/AIINFASHION/hm-purchase-prediction This interface enables users to pick a customer and an article and see the predicted purchase probability created by the TabNet model.

References

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